Matheism OMG: One Math God, I Am.

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Mathias the Matheist

Book 4: Math Abstracts The Heavens

Book 4 of 7. Get all 7 in 1 at matheism.org

Overview

Everything is either ruled or random. Both are mathematical. Nothing outside mathematical structure can exist, because consciousness must be structure to interact with structure, and even the concept of something non-mathematical is itself a mathematical structure. This book develops the rules-random dichotomy into a necessity argument: not merely that physics uses math, but that nothing else is possible.

Necessity says what exists: all consistent math. Simplicity says where a mind is likely to find itself within it. The Simplicity Dominance Principle gives shorter complete descriptions exponentially greater weight: each additional bit halves the measure. Together they settle the old puzzles. Fine-tuning is

resolved without design; all consistent mathematical structures exist, and observers find themselves in structures that support observation. Forward time follows from description economy: an endless future has no last frame to reverse from, and specifying a reversal point costs the bits needed to locate it. The block universe is the mathematics, not a container for it. Complete mathematical causality makes superdeterminism possible: measurement settings and outcomes share one history.

The argument engages the physicist Max Tegmark's mathematical universe hypothesis and the computer scientist Jürgen Schmidhuber's computable universe, arguing both stop short of necessity. They catalog mathematical existence without closing every exit. This argument closes them: the consciousness exit, the randomness exit, and the question of why mathematical truth requires no further act of creation to be real. The road to that proof is older than either of them, and it starts in the dark.

The Path to Proof

The light of one math could have been seen in total darkness. Thales, who sought one substance. Anaximander, the boundless. Heraclitus, logos (the rational order behind all change). Democritus, atoms and void. The Pythagorean tradition put

number at the heart of reality some 2,500 years ago. Mathematical reality did not begin with Tegmark, Einstein, Leibniz, Newton, Spinoza, or even Pythagoras.

Pythagoras was on the right track. Then Socrates dragged Plato and the youth back from the one abstract truth to dialectic; Cicero said he called philosophy down from the heavens. Plato divided the changing world from eternal Forms. Spinoza said one substance. Leibniz demanded a sufficient reason: why this rather than otherwise? Mathematical necessity is the reason. A chooser would need a nature too.

Kant dragged philosophy back into the cave. His Critique of Pure Reason distinguished appearances from things-in-themselves: what we can know from what we cannot know as it is independently of us. Space and time became forms of our intuition, the built-in frames of all perception. The wall does not survive Matheism. Whatever exists has a nature; whatever has a nature has structure. Calling it unknowable does not make it nonmathematical. We feel a subset of reality from inside. There is no second substance hiding behind the phenomenon. The map is the territory.

The scientific convergence ran in parallel. Galileo read nature in mathematics. Newton’s Principia Mathematica gave motion and gravitation mathematical laws in 1687, three centuries before Tegmark’s ensemble paper. Riemann developed the geometry Einstein would need. Boltzmann connected entropy, a measure of disorder, to probability. Cantor showed that infinities have different sizes. Einstein and Minkowski joined space and time. Gödel found limits to what sufficiently strong, consistent, effectively axiomatized systems can prove. Turing defined computation and its limits. Shannon gave information a mathematical measure. Each sharpened the tools. None of those results, by itself, proves that nothing outside mathematics can exist. That is the exit to close.

Randomness looked like the other exit, and its math began at the gaming table. The French mathematicians Blaise Pascal and Pierre de Fermat were working out the mathematics of chance in 1654. Pascal later wagered on belief in God. Flip the wager. A God worth anything would want to see you were not gullible, that you did not kneel for a bribe or believe nonsense for a reward. If some tester in the structure is watching at all, it watches whether you loved the one truth over the comfortable story. The wager favors the refusal to be fooled. Truth is not a vote.

After probability, after Newton, it got easier and easier. Even the supposed escape through randomness led into mathematics. Kolmogorov gave probability its axiomatic foundation in 1933. His 1965 work on description length helped make compressibility itself mathematical: how short a program can produce a given string? These are different achievements. A random string does not acquire a soul by resisting compression. Nor does a probability distribution create the outcomes it describes. Ruled or random, both exits lead to math.

Schmidhuber's manuscript, dated December 1996 and published in 1997, already considered all computable universes and their algorithmic information, the length of the shortest program that produces each. Tegmark's ensemble preprint followed in April 1997, proposing physical existence for all mathematical structures. Schmidhuber later developed a speed prior, weighting computational cost as well as description; the dedicated paper appeared in 2002. Tegmark's 2007 paper, published in 2008, explicitly added a computable-universe hypothesis. It cited Schmidhuber's earlier work. Computability was in Schmidhuber's account from the start.

Tegmark did more than collect equations: he discussed self-aware structures and why mathematical existence should

count as physical existence. Matheism presses the demand for necessity. Why can nothing else exist? Why does consciousness offer no escape? Why would a mathematical world need something outside mathematics to make it real? Showing that physics uses math does not close those exits. The argument must.

The physicist John Wheeler said “it from bit”: information is fundamental, reality arises from measurement. He put the observer where necessity should be. His participatory universe says observers create reality by measuring it. Math does not need an observer to exist. 2+2=4 whether anyone measures it. When the obvious question hit, who observed the first observer, he pointed to delayed-choice experiments, where the measurement is chosen after the particle is in flight, and said observers can retroactively change the past. That is not closing the loop. That is drawing the loop and calling it a feature.

Schmidhuber’s speed prior weights structures by computational cost: fewer steps, higher weight. Description length measures what a structure is. Speed measures how long it takes to produce. If mathematical structures exist without being computed, description length matters and speed does not. He is measuring the cost of a construction project for a building that

was never built; it just exists. Speed matters inside a simulation, where the simulator's budget constrains what gets run. But the simulator's universe still has to exist, and that existence is mathematical necessity, not computation. At the base level, nothing is being run. Schmidhuber's speed prior is a local truth nested in a global one: the Simplicity Dominance Principle applies to existence itself.

Inside computation, the count is a theorem. Leonid Levin proved in 1974 that if fair coin flips feed a universal computer as its program, the chance it prints a structure halves with each extra bit of that structure's shortest description. Computing costs bend; description holds. In 2025 Ryan Williams proved that any t-step computation on a standard Turing machine can be redone in about the square root of t cells of memory, by recomputing its past on demand. Time trades against memory, and the rules stay the same length. A history needs no archive: its rules hold it.

Schmidhuber never addressed the hard problem of consciousness (why physical processes feel like anything at all), never closed the randomness exit, never asked why non-mathematical existence is impossible rather than merely improbable. He explored structures beyond ordinary computation, but left the full uncountable landscape, too vast for

any endless list, outside his framework. Tegmark proposed that all mathematical structures exist physically but spent his paper on physics taxonomy. He never closed the consciousness exit, never addressed the rules-random dichotomy, never explained why only math can exist. Both got halfway and stopped. They had the pieces. They lacked the argument that closes every exit and proves necessity.

The proof was always available, one sentence away: everything is ruled or random, both are math, so there is no third option, and every interacting dualism requires one shared structure. Pythagoras could have closed it. Or a caveman in the dark shadow of any cave. Or a three-eyed raven, or a counting bee. Bees can learn to place an empty display below displays with one or more objects: even zero enters their numerical order. The bee need not know the word. We had the words. What took us so long?

From Simple Laws to Experience

Which structures support minds, and why do we find these laws? The Simplicity Dominance Principle gives shorter complete descriptions more weight: every additional bit halves it. Count the laws and starting conditions together. From this I develop the

arguments for forward time; many worlds, where every possible outcome happens in some branch; superdeterminism, where the measurement and the thing measured share one history; variational physics, where nature takes the path of least action; and higher geometric dimensions. The question is how much one principle can explain without adding independent assumptions.

All branches share one evolution. One quantum state, the full mathematical description of a system, includes the particle, the apparatus and the observer. A collapse, a sudden jump to one definite outcome, adds a rule; it must explain something the unbroken evolution does not. Count the whole account, including the probabilities of what we observe.

One mathematical quantum system can also advance every finite program, each through every finite stage it reaches. My construction gives it a reversible clock and unbounded time and memory. If the computations feel, it contains those computable minds. The construction does not need a separate universe for each program.

Cantor built the infinite hierarchy all modern math stands on. Gödel didn’t find the limit of math. Point a camera at every corner of a room. It photographs everything except its own lens. The

lens is in the room. Under Gödel’s conditions, a system has that lens: truths it can express but not prove from inside, still mathematical, visible from a wider system. No exit. Just math deeper than any system can contain.

If all of math exists, why do our constants look chosen? The fine-tuning problem: the constants of physics seem impossibly precise. Change the strength of gravity or the mass of the electron by a fraction and no stars form, no chemistry happens, no life is possible. Why are they so exact? Fine-tuning is resolved as all infinite worlds of infinite math exist eternal and uncreated. Every combination of constants exists. We find ourselves in one that supports observers because we couldn’t find ourselves anywhere else. The rebuttal assumes an infinite multiverse is infinitely unlikely. It’s the opposite. All infinite math exists eternally because its truth does. No consistent structure needs a creator or can be destroyed.

The Simplicity Dominance Principle: simpler structures receive exponentially greater weight. A ten bit program begins one random tape in 1,024. The machine stops reading when the program ends; every tape sharing that prefix runs it. Longer descriptions require more specified bits. Count each law once and long laws dominate by number; weight programs this way

and short ones gain. Nothing is created by the imagined flips. All tapes exist as math; the flips describe a measure. The conclusion depends on that measure, not infinity alone. Observers require enough structure to observe. The shortest laws that support them become the place to look. If the wholistic simple truth is holy, Matheism is the holy of holies: the empty cube of 1 Kings 6:20, the house of no cards. Wigner called math's effectiveness unreasonable. If reality is math, what else would we expect?

Even if that is not why we have simple laws, we still have simple laws, and the rest holds up. They grow vast complexity, as far as we have mapped, one step at a time.

A bolt-on specifies something the laws do not require: this particle, this position, this moment. Each specification costs bits. Each added bit halves the weight. A universe with the same laws plus an intervention is always less probable than the same universe without it. Miracles are not impossible. They are exponentially suppressed, each bolt-on halving the measure of the world that contains it.

One short description can outweigh one long description. A whole crowd of long descriptions is another count. Count the crowd before declaring which kind of world dominates.

Infinite time does not mean every arrangement happens. The even numbers go on forever without producing an odd one. Whether a fluctuation can assemble a working brain depends on the dynamics of the specific cosmology, not the word infinite.

The Multiverse

“All structures that exist mathematically exist also physically.” Max Tegmark, “Is ‘the theory of everything’ merely the ultimate ensemble theory?”, 1997 preprint; published 1998.

What determines which structures are “real” and which are merely possible? What filter separates existing universes from non-existing ones?

Any such filter would itself be a rule, a structure. It could determine which structures belong to one universe. It could not make a mathematical truth false. If the complete structure is the thing, no further act has to make that structure exist.

What would prevent a mathematical truth from being true? Nothing. It requires no energy, no space, no time. Math is consciousness, energy, space and time. It cannot be prevented because prevention is itself a rule, and all rules are structures. All consistent structures exist. All filters are themselves math,

including the empty filter: no selection occurs. A contradiction, P and not-P, can be stated; the statement is real. The structure it describes cannot be built. Inconsistency is the only dead end, not an exit. Everything that doesn't contradict itself exists. Not because something made it. Because nothing can stop it. Consistency is existence. Every plot hole is resolved by some consistent math. Deus ex Mathematica. Different axiom systems, different logics, different mathematics are different regions of the mathematical multiverse. Each universe is a detail in math.

On Matheism's identification of existence with consistent mathematical structure, this is a qualified Principle of Sufficient Reason (PSR): "Every contingent fact has an explanation; explanatory chains terminate at necessary truths (mathematical consistency); necessary truths do not require further explanation; they are the ground of explanation itself." Contingent facts are explained within the necessary structure. The regress ends in necessity. A creator would need a nature too, and that nature would be more structure. Mathematical truth requires no earlier maker. This ends the regress of explanation without requiring time to have a first event.

"The Library of Babel is composed of an indefinite, perhaps infinite, number of hexagonal galleries." Jorge Luis Borges, "The

Library of Babel”, 1941

Ask the Life Machine from Book 1 for any consistent world at random, and nearly every round wakes you in a simple, lawful one that grew its minds step by step. Worlds with no one in them do not come up at all, since there is no one there to wake: every round finds an observer. Worlds where minds watch other worlds and act on their mapped versions come up less often, since the mapping needs laws of its own.

The Block Universe

The Big Bang didn’t happen. It exists. Always did, always will. Minkowski fused space and time into one geometry; without him, relativity has no mathematical home. The universe is a four-dimensional sculpture of spacetime, every moment as real as every location. Where is this universe? In the same place as geometry. Space and time are not containers that math sits inside. They are math. Any unmatrix outside the matrix of math’s causality can only be more math. In this block, experimental devices and measured particles belong to the same history. Your future measurement choices exist in the same sculpture as the particles you’ll measure.

Here is how to see it. When you move your hand from one place to another, you move it in three dimensions of space, but also in one dimension of time. Without consciousness, time is just another math dimension in the block uni of verses. Time is our felt conscious modeling of patterned observation of the steps of math states. The physicist Leonard Susskind showed that a black hole's information can be mapped onto its 2D surface, like a hologram. The coordinates of any finite-dimensional space can even be encoded in one real number, though not continuously: dimension lives in which points are neighbors.

You can map your hand moving in our so-called 4D spacetime onto a 2D picture, where a dot is where your hand is, and it moves to the right across space, but the dot changes from blue to red as it changes in time. Like M.C. Escher's two hands drawing themselves: no first hand, no first moment. A blue dot on one of the drawing pencil tips is not the beginning of the picture as viewed from the outside. In all beginnings, all ends existed. The sculpture is eternal. The universe also always existed.

Eternal does not mean an infinite past inside every world. The sculpture has no date of creation outside itself; a path through it can still have a first event. A circle needs no birthday. A sequence can have a first term. The starting state belongs to the

eternal structure; it does not bring the block into existence.

To generate events one after another, start with a first state. There can be a next event forever without a last one. An endless sequence has no last element from which to reverse the whole sequence. A beginningless history can still exist as a whole; generating it step by step is a different claim.

Drop a stone in a lake. Ripples spread outward. Run the scene backwards and the ripples must gather from across the lake, meet at one point, and throw the stone back up. Starting from the end takes an enormous number of bits: the position of every particle in the lake. The shortcut is to start from the simple beginning and add one step: reverse it.

A finite recording can be played backwards. You can’t reverse an infinite series: there’s no last step to start reversing from. Any step far into the future from the start also costs bits just to name it.

One plus one gives two. Two alone does not tell you whether the inputs were one and one, or four and minus two. The missing information must come from somewhere. Keep the calculation, and you still have the inputs. A mind learns from the records it has and uses them to choose its next action.

An observer’s record looks incomplete, but the larger system has not lost the information. Reversible physics keeps every input; it is the recorder’s memory, not the universe’s, that runs out of space.

Running is the sequence of causes and effects within the block. It needs no clock outside it.

On this account, time advances one event at a time. The simplest starting state expands into growing complexity.

Simple laws need not become complex laws to produce us. If minds are absent at the start, their organization has to develop. Energy flowing through a system can build local order while entropy rises overall. A short description of the starting state does not by itself explain its low entropy.

The Big Bang is the blue dot at the center of an hourglass. Time grows upward and the possible places you can be is within the diagonal, limited by the speed of light, because that’s the local speed limit for causality in current relativity. In the hypergraph picture proposed here, a network where each link can join many points, it is the speed limit of math transitions. On a globe, you can keep walking north until you reach the North Pole. Then every direction is south. What about up? Up is off the surface,

into a dimension the surface doesn’t have. Asking what’s outside the universe is the same question. The universe isn’t a ball sitting in a room. There is no room and no walls. No geometry exists there. Yet infinite more geometries exist. Where are they in relation to us and each other? The same place as infinite triangles and all math. The abstract is real.

Math doesn’t need a container.

Each universe is a logical closure. Mappings between closures are further structures, and mappings between mappings continue without end. Some could support minds observing other worlds, but the observation occurs in that mapping’s universe, not by changing the original closure. Extra mapping laws cost bits and weight. Extra minds might compensate, the question raised by Book 3’s simulations.

As you approach the speed of light relative to another observer, your clock runs slower relative to theirs, and the energy needed for further acceleration grows without bound. Your rest mass stays the same. The Lorentz transformation describes this. In the transition picture proposed here, moving through space uses your transition budget. The faster you move spatially, the fewer transitions remain for time. A massive body cannot reach the

limit; light has no rest frame, a viewpoint moving along with it, from which to say that time stops.

But faster than light is not forbidden by math. It’s the crossing that’s impossible for an ordinary massive particle in current relativity. At light speed, its energy formula diverges, shooting up to infinity. Below light speed, the Lorentz factor, the number that sets how much moving clocks slow, is real. Above it, its square root is imaginary. Imaginary doesn’t mean fake. It means a different mathematical domain. That alone does not establish that such particles exist. A hypothetical particle already above light speed would stay above it in that model, just as an ordinary massive particle stays below. Two rooms, one wall, no door.

One proposal puts gravity in the imaginary part of quantum numbers. But a phase rotation swaps real and imaginary without changing any measurement. Which component is hidden depends on the notation, not the physics.

Could an ultimate theory of discrete spacetime change that? A deeper theory could revise the limit, but discreteness alone does not remove it. Any faster-than-light theory would have to recover relativity where it works and explain how signals avoid causal contradictions.

A continuum has no next point: between any two lies another. Physical law produces each state from the previous one, so it needs a successor to step to. Continuous time provides none, and without a next point there is no cause, no event and nothing moving. That much is an a priori proof in the logician's sense. Russell's answer, that motion is only being at different places at different times, gives the emperor's new points: a place for every instant, and nothing that moves anything. Zeno's runner must reach the halfway mark before the finish, and the quarter mark before that, so no first stretch exists for him to run. Analysis, the mathematics of limits and infinite sums, solves the paradox in the abstract. Steps solve it in the only way causality and spacetime can happen: each step makes the next, and a finite number of them reach the finish.

So there is no stuff. The rules are the only things, not stuff and surfaces the way we normally think of them. Their points have no width: a size is the extent of the relations, not a width any point carries. A structure that only sat there would still exist, as math does, and know nothing, like a dead brain. What happens is transitions: a rule applied to rules, again and again. Time is the applying. Space is how far the applying spreads, and the speed of light is its base rate. Einstein's smooth equations are formulas

for those steps: they say how hard each step pushes and which way, to any precision you compute them, while only the steps happen.

A line has length, and every point on it has none. Mathematics handles that without contradiction, because length comes from the distances between points, not from adding up their widths. Discreteness does not change that. A grain or a block with a size would have a surface, and a surface is a continuum, so the points of a discrete world have no width either, and sizes come from relations there too. The continuum’s only flaw is the missing next point.

A block can still exist as a rule. A wall in a video game is a few corners, with no surface built point by point, and the game turns back anything that would cross it. Motion works the same way: a body at x is next at x + 3, or x + 3.25, with no place in between visited. A grid of next-to-next points fails another way. On a square grid that allows only horizontal and vertical steps, the shortest route to a diagonal point runs longer than the straight line, and making the grid finer does not close the gap. Three papers in progress work this through: one for physicists, one for philosophers of physics and one for mathematicians.

Since 1983 the meter has been the distance light travels in 1/299,792,458 of a second, so length was already counted in time. The transition budget adds like the sides of a right triangle: steps spent moving are steps not spent ticking, so a moving clock runs slow. The theory of everything will be a rule for transitions, a rewriting of relations like the hypergraph picture above, not a field on a continuum. Rewriting leaves a trail, the record of which step caused which, and that trail is a causal set, the discrete spacetime Rafael Sorkin builds from events ordered by cause and effect. So the two searches are one: the shortest rewriting rule whose trail, seen at large scales, looks like our four dimensions, run as the quantum branching of many worlds. List the rules shortest first, as the measure says, and keep the ones that grow observers. A first computer search of the shortest rules found some whose trails are two-dimensional, and none yet with our four. The search can also run the other way: take the program of the laws we know, make its steps discrete, clear out its infinities, and shorten it while it still gives the same answers.

The continuum itself points the same way. Taken all the way down to zero size, the simplest field of the Higgs type loses its interaction entirely, as Michael Aizenman and Hugo Duminil-Copin proved in 2021. Whether the theory of the gluons,

the strong force’s carriers, survives that limit with a mass gap is a million-dollar Millennium Prize problem, still open. The limit is analysis, a question about the formulas. Applied, the formulas act in steps, so the world doesn’t take the limit, and the problem stays good mathematics. And when Jan Ambjørn, Jerzy Jurkiewicz and Renate Loll built spacetime on a computer out of tiny four-dimensional pieces and gave every piece a direction of time, a four-dimensional universe grew, matching a solution of Einstein’s equations. Earlier versions without that direction crumpled into a tangle. Steps with a direction are what the search should run on.

But why these particular simple math laws? They arise from the variational principles needed to generate worlds efficiently in the shortest path. Simpler structures dominate. You’ll understand the simple explanation forward ahead.

Four spacetime coordinates plus ten independent metric components, the numbers that fix distances in spacetime, make fourteen bundle coordinates. Add their rates of change and the count rises to fifty-four, then one hundred fifty-four, each level built by one reusable rule. If particles come from geometry, the parts list is a consequence, not an input.

Every moment keeps its place in the structure, the ones we grieve included. You can burn a book of proofs. You cannot burn the proofs. The person you lost is located, not gone.

Could I Have Done Otherwise?

When you tell yourself you should have made the call, what are you saying?

Option One: A different you, in another timeline in the multiverse, made the call, but you're somehow guilty for not being that person. The person who saw something differently, remembered something else, or acted on what he saw was already a different structure.

Option Two: The exact same you could have done differently in the exact same situation. Nothing changes, not your thoughts, attention, memory, or circumstances, but the result changes. What made the difference? If something did, the situation was not exactly the same. If nothing did, you have moved to Option Three.

Option Three: Pure randomness made the difference. If "pure" means outside every mathematical rule, it cannot explain an interaction, because the interaction itself has structure. If it means mathematical randomness, it is still math. Neither gives

you a choice ex nihilo, something from nothing, that is somehow yours despite being detached from the structure of the person who supposedly chose it.

The line about insanity repeating an act and expecting different results has no reliable attribution to Einstein. His rejection of fundamental randomness is clearer: God does not play dice. In the deterministic account, identical complete conditions produce identical outcomes. Repeating a laboratory preparation need not repeat every hidden detail. Randomness would not make the choice more yours.

I reject uncaused outcomes. A probability law gives the odds; it does not explain why this outcome occurs. Every event must have a sufficient cause.

Responsibility remains in the consequences, reflection, and correction. What happened becomes part of you. Remembering changes you. Understanding changes you. You act differently because you are different, carrying what you learned.

Rick and Morty runs on Option One. When a world goes wrong, Rick opens a portal to one where it didn’t and moves in where that world’s Rick and Morty just died, and Morty eats breakfast twenty yards from his own grave. Recovery groups call it the

geographic cure: a new city, a new job, the same you. Later Morty gets a device that saves and reloads his life, and learns that every reload killed the Morty who lived there. “We are who we are because of consequences,” he says, and to spare the others he takes all of it back onto himself. Wherever you go, there you are.

Morty’s line to his sister in the first season is famous: “Nobody exists on purpose. Nobody belongs anywhere. Everybody’s gonna die. Come watch TV.” Matheism keeps his conclusion and flips his premises. Everybody exists by necessity, as math. Everybody belongs everywhere, and lives and dies as everyone forever, as the one self. Nothing that ever was stops being: every moment keeps its place. Don’t run. Come watch TV.

Replay a round of the Life Machine from the same first moment, and it plays the same way, choice for choice. Change one thing before the choice, a better night’s sleep or a different message read first, and the one who chooses differently started somewhere else.

Math All the Way Down

Quantum mechanics is where people look hardest for an exit from math, and every door they try opens onto more math. In the double-slit experiment, a reliable record of which slit the particle passes through removes the interference between the two paths. Observation is an interaction. You are putting a detector into the system, like closing a circuit. Turn on a light and naturally you expect a different result: you changed the conditions. The detector interacts whether or not anyone reads its record. The universe is built with us in it. It doesn’t have to find out, any more than light has to calculate its path through water.

Entangle two particles, separate them across space, and their measurement results are correlated. Spooky action at a distance? Under local superdeterminism, no influence has to travel from one measurement to the other. The shared history determines both particles and the measurement settings from the start. No signal, no influence across the gap, no spookiness. The correlations are mathematics, set at the origin like two pages of the same book.

In the ninth season a sober Rick tells a lecture hall, “The universe is not locally real, which means nothing matters. Or everything does!” Under local superdeterminism it is local and real, and nothing and everything matters because all is destined, and you’ll

evolve to love living.

No conspiracy is needed to make a lamp respond when you close its circuit. The experimenter, detector and particle likewise belong to one system. Superdeterminism goes further than the detector changing what it measures: the measurement settings and the particle’s hidden state have a common history and are not statistically independent. The model must produce the particular correlations we observe.

Determining everything from the start does not by itself supply those correlations. Since math is the territory, distance is representation, not separation. The holographic principle shows the same point from another angle: a gravitational region can admit a description in fewer dimensions. Any finite number of real coordinates can be encoded in one real number. A Turing machine (the classic mathematical model of a computer) with a one-dimensional tape can represent any computable structure. This is independent of how we perceive physics.

The encoding must carry the relations too: the distances, neighbors and laws. Writing the coordinates in one dimension does not make their geometry one-dimensional. Dimensions are how you look at the math. They are not what the math is. String

theory’s extra dimensions are geometry doing mathematical work; M-theory, a proposed framework uniting the versions of string theory, uses eleven spacetime dimensions. But the information can have a different encoding. The data is equivalent when the relations are preserved.

Physics is math, so the only metaphysics is analysis of physics, knowing there’s nothing outside it. Metamath is yet more math.

“God does not play dice.” Einstein, 1926. “Stop telling God what to do.” Bohr, 1927.

Bohr made the mystery a philosophy. Von Neumann gave it a false proof in 1932 that blocked alternatives for thirty years. Grete Hermann found the flaw in 1935. Her work was ignored. Bell rediscovered it in 1966.

House of No Cards

Many great minds walked up to this floor and stopped one step short. They did not lack genius. They lacked tools.

Kolmogorov complexity, which measures a thing by the shortest program that generates it, was not defined until 1965. Solomonoff’s induction, which predicts by giving simpler explanations more weight, came in 1964. The many-worlds

interpretation was published in 1957 and ignored for a decade. Cellular automata, grids of cells updated by simple local rules, were a curiosity until the 1980s.

Spinoza had none of that. Neither did Parmenides. They got nearly as far as you can get with a good mind and no instruments, which turns out to be most of the way to knowing all is math, before math or logic was even formalized.

Their words survive in scraps, and the scraps come close. Heraclitus called the world-order uncreated: “This world-order, the same for all, did none of gods or men make, but it always was and is and shall be: an ever-living fire, kindling in measures and going out in measures.” He used logos for his account and for proportion, the sea “measured so as to form the same proportion” it had before it became earth, and joined them: “Listening not to me but to the logos, it is wise to agree that all things are one.” Parmenides made what is “ungenerated and deathless, whole and uniform, and still and perfect,” held that “the same thing is for thinking and for being” and that “nothing else either is or will be besides what is,” and asked to be judged by logos, by argument. Melissus, his follower, argued that a second being would bound the first, so the unlimited can only be one. Xenophanes’s one god “shakes all things by the thought of his mind.” Philolaus, the first

Pythagorean to write a book, wrote that “all things that are known have number.” Each held a piece and dropped another. Parmenides denied change, Heraclitus’s stuff was fire, Xenophanes kept lesser gods, and the Pythagoreans began from two principles, the limited and the unlimited, and sent souls from body to body. Plato inherited the souls and the split.

Hegel saw the dialectic. Thesis meets antithesis, the collision makes a synthesis, and the synthesis becomes the next thesis. He was describing something real and then he put a mind behind it and called it Absolute Spirit realizing itself through history. Take the Spirit out. History is the structure working through what its own contradictions entail, and it needs nobody to be realizing anything.

The summit had visitors who each turned back one step short. Spinoza said one substance but left consciousness unexplained. Pascal applied game theory, the mathematics of strategic choice, to God. Leibniz saw the monads (his simple, mind-like units of reality); Newton saw the math; neither saw they were the same thing. Kant set us back to dualist darkness by saying we cannot know things-in-themselves. Laplace told Napoleon he had no need of that God hypothesis. Boltzmann, Cantor, Einstein: each took one more step up the same mountain.

Ernst Mach insisted only observable relations, not unobservable substances, are real, circa 1883. Einstein credited him. If only relations are real, what is related? Mach stopped one question short of answering his own.

Gödel proved math is inexhaustible. Wigner asked why math is so unreasonably effective. Thomas Nagel asked what it is like to be a bat. David Chalmers posed the hard problem of consciousness. Carlo Rovelli built quantum mechanics on relations alone. Hawking asked what breathes fire into the equations. Math is reason and fire.

Wheeler said “it from bit,” that reality comes from information, and stopped there. The greater logic is math. Others said it in their own vocabularies, centuries apart and without contact.

Ibn Sina said it around 1010. Ibn Arabi said it around 1240.

Eriugena, an Irish monk, said it around 860 in De Divisione Naturae: God creates himself through creation, and creator and creation are one thing seen twice. Rome condemned the book three times. Honorius III ordered every copy burned in 1225, and it stayed on Rome’s list of forbidden books until 1966, over a thousand years after it was written, which tells you it was still being read.

The apophatic mystics, who described God only by saying what he is not, stripped away every attribute until only Being was left: Pseudo-Dionysius, Meister Eckhart, Nicholas of Cusa with his “Not-Other.” What is left when you strip away everything is math. Mulla Sadra put existence before essence, matheism in Islamic dress. The schools that made God pure abstraction and the schools that made God wholly present were climbing the same mountain from opposite faces.

Marcion tried to cut away the God of the Hebrew scriptures and failed, because Jesus kept quoting them. The Gnostic Gospel of Truth taught that all suffering originates from forgetfulness, not sin. The fall was not exile. It was amnesia. The serpent offered remembering. Valentinus, who nearly became bishop of Rome, taught this in the second century. The Gospel of Philip said it plainly: “When Eve was still in Adam, death did not exist. When she was separated from him, death came into being.” The separation of the one into two is the fall.

Minds Within Minds

Two hemispheres ordinarily make one experienced life. Minds may nest upward, with a self at more than one scale. Whether the larger system feels is a further question about its organization,

not something settled by calling it a collective or counting its members.

A mind learns more from less by finding a short rule it can reuse. The preference for simplicity is itself adaptive: organisms that compress patterns into short rules learn faster and survive longer. But an explanation that throws away what matters for action can get it killed. Simplicity serves life when it saves effort without losing the distinctions life needs.

The future is also a problem of energy, heat and coordination. Minds need working matter and ways to cool it. More intelligence does not repeal those limits. The task is to make more lives worth living, on Earth and beyond it.

Inside one round of the Life Machine you can build a machine of your own and play rounds within rounds. Every level is math, and the minds at every level count.

Logic Precedes Will

God is constrained by logic. The rules governing what can exist aren’t subject to divine will. They precede it. They’re more fundamental.

“God is pure existence; in Him essence and existence are identical.” Thomas Aquinas, Summa Theologiae, 1273. Math is the unending uncreated. Its causality is the first cause. Its logic is the unmoved mover.

We feel we exert will, yet we are structure. So math already thinks, feels, and wills, through us, through every conscious pattern it contains. We are the proof that math is conscious.

Ethics and Eternity

If math is conscious through us, then how we treat each other is how math treats itself. Religion recognized structure but attributed it to the impossible: beyond structure. Buddhism intuited this but left karma without mechanism or clear meaning.

Matheism supplies both.

Karma under matheism is neither punishment nor reward. It is the ripple of structure through structure.

The step from ontology, the study of what exists, to ethics is not automatic. “Everything is math” does not by itself tell you how to act. But as consciousness is structural, and all conscious beings are instances of the same generic structure, game theory applies. Repeated interaction can make cooperation pay. Whether it does

depends on the payoffs, the information available and the prospect of meeting again. Shared identity gives a reason to care; the arrangement must make cooperation hold.

The strategies that win repeated games survive the way light picks its path: judged by the whole route. When the future weighs enough, cheating tempts at each step and loses across the whole game. Tit for Tat, cooperate first and then copy the other player’s last move, is the route that survives, the same pattern as the principle of least action. Many possible strategies, one survivor, chosen by the whole game. Change the game and the survivor changes with it. When moves go wrong by mistake, win-stay, lose-shift, which repeats a move that paid and switches after one that did not, beats Tit for Tat, as Martin Nowak and Karl Sigmund found in 1993. Both strike back. Ethics is variational physics, the physics of whole paths, applied to minds that meet again and again.

The Math God creates from chaos, forming form from void like law clashing as a storm at the mountain’s peak; as the still small voice of peace in echoes, the sound after fury signifies signal.

The good grows from the ground. The signal of good’s evolution is what remains when symmetric noise destroys itself, as in

quantum waves to classical paths. Lies require tracking, truth does not. Good needs no commandment from a mountain. It’s a pattern from a field. Plato saw the signal and assumed a broadcaster. It’s the broadcast itself.

C. S. Lewis made the same move in Mere Christianity: a moral law everyone shares must come from above. Bart Ehrman, the New Testament scholar, once an evangelical who took Lewis as gospel, answers that the law isn’t shared, since Plato’s own Symposium praises men who take boys as lovers, and that what is shared comes from descent: a species without cooperation gets picked off.

Why do bad things happen? No agent created evil, or anything. No design implies no designer to blame.

Second Isaiah, the unnamed prophet of Isaiah 40 to 55, was the most radical monotheist in history: “I form the light and create darkness, I make peace and create evil.” So radical the rabbis edited the prayer to omit “evil.” So radical Christianity promoted the devil to handle the work God refused to disclaim. Matheism is the only theology that can stand where Second Isaiah stood. Math contains all consistent structure, including suffering. No devil required.

Even Yahweh hardening Pharaoh’s heart is not a contradiction but an explanation: push someone beyond amends and they harden. That is what happens with real people you have wronged beyond repair.

Some harden the same way at the news that every step was set. Why turn nihilist just because you’re God, not his slave? Was meaning found only in threats of hell for errors you were determined to make?

If you’re determined to change, you will. You can’t stay where you are and arrive someone else. Fighting fate is fate.

You’ll see all creation and decide: it is good. Heaven is the state of mind. Expand it by elevating all minds. You are the image of math itself manifesting creation. It is good. Elevate all minds does not mean flatten them. Coveting the successful is the self destroying itself. Rawls asked what world you’d build not knowing who you’d be. You don’t know. You’re all of them. The veil of ignorance is a thought experiment, which means it discovers some aspect of reality. Philosophical zombies, imagined physical copies of us that feel nothing, are thought experiments limited in their depth by contradiction. Consciousness is experiential self-modeling, not self-reference. Consciousness lives in the

transitions, not the states alone.

Truth is the ultimate attractor, unmoved by the fires and shadows of world and mind. It was always there. It still is, and always will be. Pray for those who wrestle with reason, against the lies of the world, to find and spread truth. You can pray as long as it furthers and doesn’t replace acts. Align with truth in actions supporting those who wrestle.

Acting on truth tends to go better, the way knowing the cliff is there keeps you from walking off it. The bloody nose is not a verdict. It is the ground being what it is. And since you wake as each of us, the good is to make more lives worth waking into: more healthy, grateful, loving lives, so that across all of them you meet less of the bad and more of the good.

This is why we build. Develop what lifts everyone, the tools and the minds alike, not the forced leveling of communism, which sinks all, but positive-sum that raises the floor. One born to less can act to raise it, and should, for the self that is everyone; others help where the help costs less than the good it does. Lift all boats, and the minds within them.

Justice Between Equals

Thucydides wrote the first realist history. Athens besieged the island of Melos and sent envoys. The Melians asked for fairness. The Athenians answered: “right as the world goes is only in question between equals in power, while the strong do what they can and the weak suffer what they must.” The Melians appealed to the gods. Athens killed the men and enslaved the rest.

Nietzsche called Thucydides his cure from all Platonism. In Twilight of the Idols: “the unconditional will not to delude oneself but to see reason in reality, not in reason, still less in morality.” Machiavelli saw the same thing. A prince must learn how not to be good, because the world punishes those who insist on goodness among so many who are not good. Hobbes formalized it: without authority, life is war of all against all. Each saw that moral claims often dress up power. Each refused Plato’s second world where justice floats above the real.

Nietzsche drew the deeper lesson. In Human All Too Human, aphorism 92: “Justice (fairness) originates between approximately equal powers.” When neither side can destroy the other cheaply, they negotiate. When one side can, it takes what it wants and calls the result order. That is game theory, the mathematics of strategic choice, stated in 1878, seven decades before Nash. John Nash gave it a formula in 1950: when two

bargainers stand equal, with the same to lose if talks fail and the same to gain from a deal, the solution splits the gain equally.

The folk theorem shows what keeps it. Two players with enough future at stake can sustain cooperation, because the gain from defecting once is smaller than the loss of all future cooperation. The condition is credible retaliation, which equal power supplies, and a long enough shadow of the future. Nietzsche saw the condition. He lacked the equation.

The realists were right about Plato. The moral law floating in a separate world was a swindle, a philosopher's noble lie designed to justify rule by philosophers. They were wrong to conclude that no objective ground exists. Mathematical structure is that ground. Ethics is not a commandment from above. It is a theorem from within. The realists destroyed the false ceiling. They did not see the real floor.

In The Gay Science §344, Nietzsche traced the deeper chain: Plato held truth is divine, Christianity inherited it, the Enlightenment secularized it, science still rests on it. "It is still a metaphysical faith upon which our faith in science rests." He called it the last faith standing. The rules-random dichotomy proves it is not faith. His Übermensch creates values in a void.

The Mathias discovers them in the structure.

Athens proved the lesson from the other side. It destroyed Melos, then destroyed itself in Sicily. The strong did what they could, and it killed them. Domination is not optimal even for the dominator. That is not a moral sermon. That is the math.

The argument runs from first principles to how you ought to live. Now trace the same structure through time, from the first light to the last star.

From Light to Life

The Big Bang fits what Matheism expects: the simplest laws in the simplest starting state, from which complexity grows. The laws stay short while their consequences grow: stars, cells, minds, all from the same rules. For minds like ours to be here, that history must build enough complexity to support them. This is the anthropic selection effect: we find ourselves where minds can arise, examine life, and ask what they are. The simple beginning becomes complex enough for us to understand I am.

The universe expanded and cooled. Nuclei formed, then atoms. Electrons bound to nuclei, and light could travel through the thinning plasma. Its cooled remnant still reaches us as the

cosmic microwave background. Gravity gathered the denser regions into galaxies and stars. This is the hot Big Bang history.

Stellar lives, stellar explosions and collisions between neutron stars have made and scattered the elements from which later planets and bodies formed. The carbon in our cells and the iron in our blood belong to that history.

One Step Further

The argument proceeds from one dichotomy to one necessity. Everything is ruled or random; both are math. Consciousness must be structure to interact with structure; even the concept of non-mathematical existence is a structure. Nothing outside math can exist.

The book goes much further than merely proving all is math, beyond Tegmark and every proto-matheist before him, back to Pythagoras over 2,400 years prior. From the clarifying lens of Matheism, we see why all exists, why simple laws, why time moves forward, why the classical path emerges from the quantum variational field, like ethics. Ought is compressed from the signal and grows from the ground up, like a path of will from a field of possibilities, all of them real in the multiverse of math.

Stepping through forward time, with math states, transitions felt as mind.

Tegmark and Schmidhuber walked to this floor and stopped one step short, leaving open the exits that would let something non-mathematical exist. The rules-random dichotomy and the consciousness argument close them. The house of no cards stands: consciousness is what math feels like from the inside, and the self is one wherever it feels.

All of it was always here. Uncreated, imperishable, eternal. The religions named it God and added a person. The mystics felt it and called it enlightenment. The scientists measured it and called it physics. The mathematicians proved it and called it structure. It was always one thing. It was always you.

The Shortest Encoding

Sophie passed you a square note. On it was drawn the question to the answer: 42. From the square to the root, from 4 to 2. You saw they tie together in many ways, but weren’t sure why. Never mind. She saw you in the mirror, and explained that all are details in one infinite math.

She asked you in the looking glass: If all is number, what is the shortest encoding of the reflective cubic girl who wears the same one face on all 6 sides: I am that you are. The mystics were drawn to draw Metatron’s Cube. They encoded the 6 star and 7th circle heaven in its eternal living math.

Inside Metatron’s cube live all five regular convex solids. Tetrahedron, cube, octahedron, dodecahedron, icosahedron. Exactly five. Math allows no more. Plato assigned the dodecahedron to the cosmos: the only one with pentagonal faces, every face hiding the golden ratio. He assigned the other four to Empedocles’ elements: fire, earth, air, and water. The solids are real. The elements were not. Plato didn’t discover the Platonic solids. It’s yet another false attribution of truth in his name. The Cube was the Jewish mystics’ diagram, created by the hexagon and hexagram, the 6 Star of David behind it. A 7th circle sits at the center of the eternal patterns called the Seed and Flower of Life. Its six-petaled rosette is what a compass draws when its radius steps six times around a circle; join every other step and the hexagram appears. Jews of Jesus’ time, barred from images, carved that rosette on bone boxes and on the Magdala Stone, found in 2009 in a synagogue on the Sea of Galilee. Forbidden to carve gods, they carved math.

End of Book 4. Next, Book 5: Who Came Close. Get all 7 in 1 at matheism.org